

Validity Analysis: FOLIO

Target context: US High School Competitive Debate — Argument Quality Assessment

Overall risk: **HIGH**

Deployment Context

We are a US debate coaching platform evaluating whether an LLM can assess the strength of informal, real-world arguments made by American high school students in competitive debate contexts. Arguments involve empirical claims, value judgments, rhetorical moves, and evidence of varying quality. The system should assess argument strength holistically — including logical validity, evidence quality, relevance, and persuasive effectiveness.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	1	Serious concern	high
Input Content ■	1	Serious concern	high
Input Form ✓	4	Minor gaps	high
Output Ontology ■	1	Serious concern	high
Output Content ■	1	Serious concern	high
Output Form ■	1	Serious concern	high
Average	1.5		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

FOLIO is a high-quality benchmark for formal deductive entailment over natural-language premises, with rigorous annotation, verified labels, and substantial logical complexity. However, for the US high school competitive debate argument-quality assessment deployment, FOLIO is fundamentally misaligned across five of six validity dimensions. The input ontology omits all deployment-required argument categories (evidence-backed claims, value-laden arguments, rhetorical moves, debate-structural arguments). The input content is deliberately stripped of normative contestation and evidentiary disputes — the very content types most central to debate. The output ontology (True/False/Unknown) cannot represent multi-dimensional sub-scores, qualitative feedback, or pairwise rankings. The output content's correctness standard (theorem-prover verification by FOL-trained CS students) is categorically different from the pluralistic,

format-varying, circuit-varying debate-community judgment standard. The output form produces a single categorical label, structurally uninformative for the deployment's required outputs. Only the input form (text in standard American English) aligns well. Dataset analysis across 57 sampled examples confirms these gaps empirically: no example contains evidence citation, normative contestation, debate-structural arguments, sub-scores, rationales, or pairwise comparisons. The benchmark provides at most a narrow upper-bound test on multi-step deductive validity, which is one minor component of the deployment's logical-structure sub-score — far from sufficient for deployment-level evaluation.

Practical Guidance

What This Benchmark Measures

FOLIO measures formal deductive entailment over natural-language premise sets and NL-to-FOL translation accuracy. Within the deployment's framework, it provides signal for only a narrow sub-component of the 'logical structure and warrant quality' sub-score — specifically, whether a model can correctly determine entailment relationships under multi-step deduction. It does not measure evidence quality (the highest-weight deployment sub-dimension), relevance/responsiveness, persuasive effectiveness, debate-structural argument recognition, value-laden argument assessment, or format/circuit-conditioned quality norms.

Construct Depth

FOLIO probes deductive validity rigorously, with reasoning depth modal at 4 and a diverse AST distribution [Q60, Q61], and provides a clean expert/non-expert performance gap demonstrating the construct is meaningfully measured [Q111]. However, the construct it probes deeply is *not* the construct the deployment needs. The benchmark provides no probing of inductive reasoning, evidence credibility, warrant-claim coherence in hedged empirical claims, or pluralistic argument quality. The depth of FOLIO's measurement on deductive validity does not compensate for the breadth gap across the deployment's required constructs.

What Else You Need

Substantial supplementation is required across nearly every dimension. For input ontology (IO) and input content (IC): supplement with debate-specific corpora such as DebateSum [WEB-21] and OpenDebateEvidence [WEB-22] for evidence-card content, and develop NSDA-format-specific argument-type taxonomies covering DAs, CPs, kritiks, T shells, theory, and impact calculus. For output ontology (OO) and output form (OF): construct a multi-dimensional rubric with separate sub-scores (evidence, logic, relevance, persuasion), qualitative rationale generation, and pairwise comparison structures, drawing on DebateBench [WEB-28], IBM ArgQ pairwise datasets [WEB-24], and CompAQA [WEB-26]. For output content (OC): recruit a panel of NSDA-credentialed judges and coaches, stratified by format (Policy/LD/PF/Congressional) and circuit (lay/tech), to provide ground-truth annotations that reflect actual debate-community norms. Pair circuit-conditioned scoring with judge-paradigm metadata from Tabroom. Verify NSDA evidence-ethics policy compliance [NSDA verification gap noted in regional context]. FOLIO can be retained only as a narrow component sub-test for deductive-validity-specific capability, not as an end-to-end deployment proxy.

ID	Evidence
Q60	"As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1." (p.5)
Q61	"Table 1 shows that FOLIO also has a much larger number of distinct ASTs than the previous datasets, indicating that FOLIO is much more logically diverse." (p.5)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
WEB-21	DebateSum evidence cards include hedged author/citation register absent from FOLIO (https://www.semanticscholar.org/paper/DebateSum:-A-large-scale-argument-mining-and-Roush-Balaji/d27b3096c0707f42b068fc9b3b47ce87560cbd56)
WEB-22	OpenDebateEvidence transcripts include the hedged, warrant-citing register typical of competitive debate (https://arxiv.org/abs/2406.14657)
WEB-24	https://research.ibm.com/blog/project-debater-api

WEB-26	CompAQA argues for combined pairwise classification and ranking — the kind of output form the deployment requires and FOLIO lacks (https://www.mdpi.com/2079-9292/13/20/4088)
WEB-28	DebateBench evaluates speech-level scoring and ranking — confirms multi-output debate benchmarks are feasible but FOLIO does not match (https://arxiv.org/pdf/2502.06279)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US High School Competitive Debate — Argument Quality Assessment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's input ontology is organized exclusively around formal deductive entailment and NL-FOL translation [Q15, Q63, Q67], with task complexity operationalized through reasoning depth and AST diversity [Q60, Q61]. The deployment requires assessment of inductive/evidentiary reasoning, value-laden claims, rhetorical moves, and debate-specific structural arguments (DAs, CPs, kritiks, T shells, theory) — none of which appear in FOLIO's taxonomy. Dataset analysis confirms zero coverage: even when 'inductive reasoning' appears, it is treated as a deductive predicate label rather than as an actual inductive inference [DATASET-D3, D25]. The HIGH-priority elicitation flag on IO is fully borne out.

ID	Evidence
Q15	"We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises." (p.1)
Q60	"As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1." (p.5)
Q61	"Table 1 shows that FOLIO also has a much larger number of distinct ASTs than the previous datasets, indicating that FOLIO is much more logically diverse." (p.5)
Q63	"We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation." (p.6)
Q67	"We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset." (p.6)
D3	All inductive reasoning processes derive general principles from a body of observations. Two major types of reasoning rules are inductive reasoning and deductive reasoning. — Story uses "inductive reasoning" as a named category but treats it deductively — no actual inductive inference is performed
D25	$\forall x (\text{MajorArgumentForm}(x) \rightarrow (\text{InductiveReasoning}(x) \oplus \text{DeductiveReasoning}(x)))$ — Inductive reasoning is modeled as a named entity/predicate; the reasoning task itself remains purely deductive

Strengths

- Provides a rigorous upper-bound test for the narrow sub-component of multi-step deductive validity, with reasoning depth modal at 4 and 28.7% of examples requiring 5+ inference steps [Q60], and substantially more distinct ASTs than prior datasets [Q61]

ID	Evidence
Q60	"As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1." (p.5)
Q61	"Table 1 shows that FOLIO also has a much larger number of distinct ASTs than the previous datasets, indicating that FOLIO is much more logically diverse." (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment

IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: The deployment requires test categories spanning: (a) evidence-backed empirical arguments, (b) value-laden/normative claims (LD, Congressional), (c) rhetorical/persuasive moves, (d) debate-specific structural arguments (DAs, CPs, kritiks, T, theory, impact calculus), and (e) pairwise rebuttal comparison. Evidence quality is identified as the highest-weight sub-dimension per elicitation.

IO-2: FOLIO's taxonomy omits all five of the above categories. The benchmark is framed strictly around deductive entailment over premise sets [Q15, Q66] and NL-FOL translation [Q18, Q67]. Dataset analysis confirms zero presence of debate-specific tokens (disadvantage, counterplan, kritik, topicality, solvency, impact, turn, burden, framework, value, criterion) across the 57 sampled examples.

IO-3: FOLIO includes the NL-FOL translation task [Q3, Q67] which is largely irrelevant to the deployment — debate coaching does not require translating speeches into formal first-order logic. Syllogism-template-instantiated stories [Q42, Q43] also have low ecological validity for actual debate discourse.

IO-4: Category gaps are total relative to deployment-critical evaluation: evidence-handling, value-laden argument assessment, rhetorical effectiveness, and debate-structural moves are entirely unrepresented. Authors themselves note prior NL benchmarks are inadequate for measuring complex reasoning [Q8, Q9, Q10], but FOLIO's response narrows the scope further into pure deductive logic rather than broadening to argumentation.

ID	Evidence
Q3	"In addition to the main NL reasoning task, NL-FOL pairs in FOLIO constitute a new NL-FOL translation dataset." (p.1)
Q8	"However, existing natural language tasks are inadequate in measuring the complex logical reasoning capability of a model (Srivastava et al., 2023; Saparov and He, 2023; Tian et al., 2021)." (p.1)
Q9	"Some of these common benchmarks do not specifically evaluate logical reasoning independently of other forms of reasoning (Yu et al., 2020; Liu et al., 2021)." (p.1)
Q10	"Those specifically designed for measuring logical reasoning are insufficient in terms of logical reasoning complexity and natural language variety." (p.1)
Q15	"We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises." (p.1)
Q18	"Based on our annotations, we propose a new NL-FOL translation task where an NL reasoning example is translated into its FOL counterpart." (p.2)
Q42	"In this approach, we first generate logically valid stories, which are templates containing abstract categories and entities, by combining multiple syllogisms into a single story template: the conclusion of one syllogism is used as a premise for the next syllogism." (p.4)
Q43	"There are 256 logically distinct types of syllogisms and 24 of them are valid (Lehman, 1973). We use various combinations of 24 valid syllogisms. We also add in conjunction, disjunction, and implication." (p.4)
Q63	"We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation." (p.6)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q67	"We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset." (p.6)
WEB-22	OpenDebateEvidence transcripts include the hedged, warrant-citing register typical of competitive debate (https://arxiv.org/abs/2406.14657)
WEB-28	DebateBench evaluates speech-level scoring and ranking — confirms multi-output debate benchmarks are feasible but FOLIO does not match (https://arxiv.org/pdf/2502.06279)

Remediation

Gap 1: FOLIO's taxonomy covers only formal deductive entailment and NL-FOL translation; the deployment requires evidentiary, value-laden, rhetorical, and debate-structural argument categories.

Recommendation: Construct a debate-specific argument-type taxonomy aligned to NSDA formats (Policy: DA/CP/T/K; LD: V/C framework, plans, kritiks, theory; PF: contention/weighing; Congressional: legislative advocacy/oratory). Use FOLIO only as a narrow sub-test for the deductive-validity component of the logical-structure sub-score, not as a primary evaluation.

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US High School Competitive Debate — Argument Quality Assessment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO content is deliberately constructed to be free of evidentiary disputes, normative contestation, and culturally embedded judgment [\[Q47, Q127\]](#), with stories rewritten to remove biases and opinionated language [\[Q48, Q127\]](#). The deployment, by contrast, centers on contested empirical claims, value-laden arguments, and miscontextualized evidence as the highest-weight sub-dimensions. Dataset analysis confirms that even when politically adjacent content appears (Russia embargo, state ownership, TikTok addictiveness, human rights), it is rendered as deductive predicate labels rather than as evidentiary or normative claims [\[DATASET-D2, D4, D5, D10, D22\]](#). The HIGH-priority elicitation flag on IC is fully borne out.

ID	Evidence
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)
Q48	"Toward these objectives, we manually screened all stories and found that 39.2% of the stories suffer from at least one of these issues. We implemented a detailed protocol to rewrite these stories." (p.4)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D2	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. ... There is an embargo on Russian foreign trade goods. — Macro-economics scenario; premises stated as given facts, not cited evidence requiring credibility assessment
D4	Everyone at the business conference is either an investor or an entrepreneur. ... Ho is at the business conference and prefers state ownership of the means of production. — Normative terms (planned economy, state ownership, ardent communist) appear only as predicate labels; no value judgment is required or possible
D5	Everyone who is entitled to national social insurance coverage can have their medical bills partially covered. ... Either Mei is a North Korean and can have medical bills partially covered, or neither is true. — Policy-adjacent content framed as pure deductive entailment; no contested empirical or value claim is assessed
D10	If someone is a seasoned software engineer interviewer at Google, then they know what breath-first-search is. ... Jack is entitled to the right to life and liberty, has human rights, or knows about the first-in-first-out data structure. — Human rights framing used as deductive predicate; no normative assessment required or possible
D22	If a social media application is addictive, then it is not ideal for preteens. TikTok is a social media application, and it is not ideal for preteens. — Contemporary tech topic; "addictive" treated as a logical predicate, not a value-laden or contested empirical claim

Strengths

- Diverse topical breadth in WikiLogic stories covers geography, science, economics, technology, and consumer products [\[Q38, Q62, DATASET-D2, D15, D16\]](#), so the benchmark does not over-narrow into a single domain vocabulary
- Authors deliberately screened out biased/stereotyped content [\[Q47, Q127\]](#) — for general logical-validity sub-testing this is a quality-positive even though it removes the very normative contestation the deployment requires

ID	Evidence
Q38	"This results in a wide range of topics, abundant AST variations, and a wide vocabulary for FOLIO." (p.4)
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)

Q62	"Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories." (p.5)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D2	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. ... There is an embargo on Russian foreign trade goods. — Macro-economics scenario; premises stated as given facts, not cited evidence requiring credibility assessment
D15	All velvet-finish lipsticks in the Rouge Dior set, Lunar New Year Limited Edition are refillable. ... ROUGE Dior Colored Lip Balm 999 is a lipstick in the set, and it either has 'rosewood' in its official description or has a velvet finish. — Real consumer product used as topic seed; content is factual/commercial, not argumentative or evidentiary
D16	All orphan planets are rogue planets. If PSO J318.5–22 is not both a rogue planet and a planet gravitationally bound by the Sun, then it is a rogue planet. — Scientific real-world entity used as topic; premises are stated as fixed facts, not contested empirical claims

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: The deployment requires content that reflects contested empirical claims (PF/Policy), value disputes (LD/Congressional), and circuit-specific argument registers (lay vs. tech). Content must include cited evidence with sources, authors, and recency [[WEB-21](#), [WEB-22](#)]. FOLIO content does not require this — premises are stated as definitional universals or given facts [[DATASET-D1](#), [D2](#), [D17](#), [D24](#)].

IC-2: FOLIO content is deliberately neutralized of normatively charged content [[Q47](#), [Q127](#)], and 'psychologically fundamental generalizations' are explicitly accepted as a workaround [[Q128](#)]. The deployment requires content that *engages* contested value claims (e.g., 'civil liberties vs. national security' per 2026 NSDA LD topics [[WEB-18](#)]) — the benchmark systematically excludes this content type.

IC-3: FOLIO does not require region-specific knowledge in a problematic way (American English, broad topic seeds), but it does not contain Western-debate-specific knowledge either — no evidence cards, no warrant-citation patterns, no spread/flow language. The mismatch is in argument register, not regional culture per se.

IC-4: INSUFFICIENT DOCUMENTATION — the paper does not report regional annotator review. The dataset analysis substitutes for this: 57 sampled examples were reviewed against deployment criteria and found to contain zero competitive-debate content.

IC-5: Content issues are severe: (a) no evidence-citation structure — confirmed across all sampled examples; (b) no hedged empirical claims [[DATASET-D24](#), [D22](#)]; (c) no value-pluralism content [[DATASET-D4](#), [D5](#)]; (d) potential pretraining contamination of WikiLogic stories [[Q104](#), [DATASET concern 9](#)].

ID	Evidence
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)
Q104	"Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data." (p.8)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)

Q128	"We accept certain classes of "psychologically fundamental generalizations" (Leslie, 2008), however, such as "Covid is transmitted through the air" or "Tigers eat other animals," that may not be factually invariant but add logical and semantic nuances to the stories." (p.13)
D1	No plants are fungi. Mushrooms are fungi. / No plants are mushrooms. — Simple two-premise syllogism; purely formal deductive structure with no evidentiary content
D2	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. ... There is an embargo on Russian foreign trade goods. — Macro-economics scenario; premises stated as given facts, not cited evidence requiring credibility assessment
D4	Everyone at the business conference is either an investor or an entrepreneur. ... Ho is at the business conference and prefers state ownership of the means of production. — Normative terms (planned economy, state ownership, ardent communist) appear only as predicate labels; no value judgment is required or possible
D5	Everyone who is entitled to national social insurance coverage can have their medical bills partially covered. ... Either Mei is a North Korean and can have medical bills partially covered, or neither is true. — Policy-adjacent content framed as pure deductive entailment; no contested empirical or value claim is assessed
D17	The world's only major large passenger aircraft manufacturers are Boeing and Airbus. ... There exists a SpaceX commercial aircraft. — Factual world knowledge embedded as premise; no evidence citation, credibility assessment, or warrant evaluation
D22	If a social media application is addictive, then it is not ideal for preteens. TikTok is a social media application, and it is not ideal for preteens. — Contemporary tech topic; "addictive" treated as a logical predicate, not a value-laden or contested empirical claim
D24	Everyone working at Meta has a high income. ... James has a car or works at Meta. — Contemporary workplace scenario; premises stated as definitional universals, not empirically disputed claims
WEB-18	NSDA topics include normatively contested resolutions (civil liberties vs. national security, plea bargaining justice, encryption lawful access) — the exact content FOLIO excludes (https://www.speechanddebate.org/topics/)
WEB-21	DebateSum evidence cards include hedged author/citation register absent from FOLIO (https://www.semanticscholar.org/paper/DebateSum:-A-large-scale-argument-mining-and-Roush-Balaji/d27b3096c0707f42b068fc9b3b47ce87560cbd56)
WEB-22	OpenDebateEvidence transcripts include the hedged, warrant-citing register typical of competitive debate (https://arxiv.org/abs/2406.14657)

Information Gaps

- No paper-level documentation of how debate-community reviewers would assess FOLIO content (substituted by dataset analysis)

Requires Expert Verification

- Whether any subset of WikiLogic stories happens to mirror specific Policy/PF resolution content closely enough to provide partial deployment signal — likely zero based on sample, but full corpus not reviewed

Remediation

Gap 1: FOLIO content deliberately excludes normative contestation, evidentiary disputes, and value-laden claims — the highest-weight content types for the deployment.

Recommendation: Supplement with content drawn from OpenDebateEvidence [WEB-22] and DebateSum [WEB-21], which provide actual US competitive debate evidence with citation/author/recency metadata. Curate format-specific test sets covering current NSDA resolutions [WEB-18] across Policy, LD, PF, and Congressional, including value-laden LD topics and empirically contested PF topics.

Gap 2: NSDA evidence-ethics rules and AI-generated content policies are deployment-critical but flagged [NEEDS VERIFICATION] in the regional context; current NSDA rulebook URLs were not retrieved.

Recommendation: Before deployment, retrieve current NSDA evidence-ethics rules and any state-level AI policies (e.g., Ohio SDA policy [WEB-30]). Build evidence-quality scoring criteria to align with NSDA standards on cutting, citing, and contextualizing evidence, and include explicit detection of AI-generated content per state-level prohibitions.

ID	Evidence
WEB-18	NSDA topics include normatively contested resolutions (civil liberties vs. national security, plea bargaining justice, encryption lawful access) — the exact content FOLIO excludes (https://www.speechanddebate.org/topics/)

WEB-21	DebateSum evidence cards include hedged author/citation register absent from FOLIO (https://www.semanticscholar.org/paper/DebateSum:-A-large-scale-argument-mining-and-Roush-Balaji/d27b3096c0707f42b068fc9b3b47ce87560cbd56)
WEB-22	OpenDebateEvidence transcripts include the hedged, warrant-citing register typical of competitive debate (https://arxiv.org/abs/2406.14657)
WEB-30	https://sites.google.com/theosda.org/theosda/resources/debate

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: FOLIO | Context: US High School Competitive Debate — Argument Quality Assessment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO is text-only, standard American English, with grammar checked via Grammarly and reviewed by English Literature annotators [Q49, Q50]. The deployment is also text-only English (with possible speech-transcript input out of current scope). Dataset analysis confirms no modality or script mismatch across all 57 sampled examples. The IF priority was correctly flagged LOWER in elicitation; main residual concern is register mismatch — FOLIO uses logic-textbook universal-statement register, while debate discourse uses hedged, evidence-citing, warrant-heavy language.

ID	Evidence
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
Q50	"Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness." (p.4)

Strengths

- Text-only English with no modality, script, or language mismatch [Q22, DATASET-D1, D8]
- Grammar and naturalness checked by linguistically trained annotators [Q49, Q50]
- Standardized NL conventions reduce ambiguity [Q130-Q133]

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
Q50	"Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness." (p.4)
Q130	"We always use "either-or" to express exclusive disjunction." (p.13)
Q131	"We use either "A or B" or "A or B, or both" to express inclusive disjunction." (p.13)
Q132	"It is more natural to say "Some A is B" rather than "there exists an A such that A is B."" (p.13)
Q133	"All A are B" can be more natural than "If A then B"." (p.13)
D1	No plants are fungi. Mushrooms are fungi. / No plants are mushrooms. — Simple two-premise syllogism; purely formal deductive structure with no evidentiary content
D8	Harry, who lives in Potterville either yells or flies. Potter, who lives in Potterville, is a wizard and flies. — Fictional fantasy premise set; content has no overlap with competitive debate argument types

Checklist

Checklist item definitions:

ID	Assessment Question
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IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both source and target are text in standard American English [Q49, Q50]. Signal distribution matches at the modality level; no resolution/sampling-rate analog applies to text.

IF-2: US school infrastructure supports text input via Chromebooks and personal devices per deployment context. No infrastructure mismatch.

IF-3: Register mismatch is a domain-specific form difference: FOLIO uses formal-universal sentences ('All X are Y') with named constants [DATASET-D1, D24], while debate discourse uses hedged citations, spread delivery transcripts, and warrant-heavy phrasing [WEB-21, WEB-22]. If speech transcripts (with disfluencies, mid-sentence corrections) enter scope, the gap widens further.

IF-4: Form mismatches at the modality level are minimal; register-level differences are documentable but secondary to the IO/IC/OO/OC/OF mismatches.

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
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D1	No plants are fungi. Mushrooms are fungi. / No plants are mushrooms. — Simple two-premise syllogism; purely formal deductive structure with no evidentiary content
D24	Everyone working at Meta has a high income. ... James has a car or works at Meta. — Contemporary workplace scenario; premises stated as definitional universals, not empirically disputed claims
WEB-21	DebateSum evidence cards include hedged author/citation register absent from FOLIO (https://www.semanticscholar.org/paper/DebateSum:-A-large-scale-argument-mining-and-Roush-Balaji/d27b3096c0707f42b068fc9b3b47ce87560cbd56)
WEB-22	OpenDebateEvidence transcripts include the hedged, warrant-citing register typical of competitive debate (https://arxiv.org/abs/2406.14657)

Information Gaps

- Whether deployment will include ASR-transcribed speech (with disfluencies, spread delivery artifacts) — out of current scope per deployment context but flagged as potential future expansion

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US High School Competitive Debate — Argument Quality Assessment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's output label space is the three-way categorical scheme True/False/Unknown [Q36, Q66], with FOL operators restricted to standard truth-functional connectives and quantifiers [Q138]. Temporal logic, modal logic, and Davidsonian semantics are explicitly excluded [Q139-Q142]. The deployment requires multi-dimensional sub-scores across evidence quality, logical structure, relevance, and persuasive effectiveness, plus qualitative coaching feedback, pairwise rebuttal rankings, and format/circuit-conditioned norms. Dataset analysis confirms every example produces exactly one categorical label with no sub-scores, no rationales, and no comparative structure [DATASET-D11, D13, D18, D23]. The HIGH-priority elicitation flag on OO is fully borne out.

ID	Evidence
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)
Q139	"Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics." (p.13)
Q140	"Consequently, they are beyond the scope of the definition of first-order logic used in our dataset." (p.13)
Q141	"We use n-place predicates when applicable for the expressivity of the FOL formulas." (p.13)
Q142	"However, we do not use the Davidsonian (Davidson, 2001) or neo-Davidsonian semantics (Parsons, 1990) because translating the majority of the FOL formulas in our dataset only requires one-place and two-place predicates." (p.13)
D11	Everything in Size Town is big or small. ... The bird is in Size Town and it is not both heavy and still. — Fully abstract fictional scenario; label is True/False/Uncertain with no sub-dimensions
D13	All professional athletes spend most of their time on sports. All Olympic gold medal winners are professional athletes. ... If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner. — Six-premise chain requiring multi-step deductive inference; output is single True/False/Unknown label
D18	No trick-shot artist in Yale's varsity team struggles with half court shots. ... Jack is on Yale's varsity team, and he is either a trick-shot artist or he successfully shoots a high percentage of 3-pointers. — Same story_id generates multiple conclusions (examples 1134, 1141, 1140) — confirms single-label per conclusion structure
D23	Xiufeng, Xiangshan, Diecai, Qixing are districts in the city of Guilin. ... Kowloon District is in Hong Kong. — Geographic factual premises; conclusion about a completely different city is Uncertain — confirms open-world assumption

Strengths

- Output label space is unambiguous and well-defined [Q135-Q137] — for the narrow sub-component of logical validity, the three-way scheme is principled
- Open-world assumption is correctly handled via the Unknown label [DATASET-D23]

ID	Evidence
Q135	"FOL enables deriving facts from other facts (Russell and Norvig, 2010)." (p.13)

Q136	"FOL, as a logical form, is a more explicit logical representation than its NL counterpart and can be used as input to an FOL prover in order to obtain the exact truth values for the conclusions." (p.13)
Q137	"FOL has no ambiguity while ambiguity can occur at various levels of NLP." (p.13)
D23	Xiufeng, Xiangshan, Diecai, Qixing are districts in the city of Guilin. ... Kowloon District is in Hong Kong. — Geographic factual premises; conclusion about a completely different city is Uncertain — confirms open-world assumption

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: FOLIO labels are True/False/Unknown [[Q36](#), [Q66](#)]. These categories do not map to any deployment-required output: sub-score per dimension, qualitative feedback, pairwise ranking, or circuit-conditioned scoring [elicitation A2, A3].

OO-2: All deployment-relevant output categories are missing: (a) evidence-quality sub-score, (b) relevance/responsiveness sub-score, (c) persuasive-effectiveness sub-score, (d) pairwise comparative ranking, (e) format/circuit-conditioned quality bands. Note that BP-debate scoring benchmarks exist [[WEB-28](#)] but FOLIO does not approximate them.

OO-3: FOLIO's output ontology encodes the assumption that argument correctness is mechanically verifiable via theorem proving [[Q2](#), [Q135-Q137](#)]. The deployment's correctness standard is pluralistic and community-norm-based — these assumptions are categorically incompatible.

OO-4: Stakeholder-driven taxonomy redesign would not be a 'redesign' of FOLIO — it would require constructing an entirely separate evaluation with no shared output space. FOLIO cannot be adapted; only supplemented.

OO-5: Taxonomy issues represent a fundamental structural validity violation. Benchmark performance metrics (accuracy over True/False/Unknown, majority baseline 38.5% [[Q94](#)]) cannot inform any of the deployment's required output dimensions.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
Q135	"FOL enables deriving facts from other facts (Russell and Norvig, 2010)." (p.13)
Q136	"FOL, as a logical form, is a more explicit logical representation than its NL counterpart and can be used as input to an FOL prover in order to obtain the exact truth values for the conclusions." (p.13)
Q137	"FOL has no ambiguity while ambiguity can occur at various levels of NLP." (p.13)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)
Q139	"Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics." (p.13)

WEB-26	CompAQA argues for combined pairwise classification and ranking — the kind of output form the deployment requires and FOLIO lacks (https://www.mdpi.com/2079-9292/13/20/4088)
WEB-28	DebateBench evaluates speech-level scoring and ranking — confirms multi-output debate benchmarks are feasible but FOLIO does not match (https://arxiv.org/pdf/2502.06279)

Remediation

Gap 1: FOLIO's True/False/Unknown label space cannot represent the deployment's multi-dimensional sub-scores, qualitative feedback, pairwise rankings, or format/circuit-conditioned norms.

Recommendation: Design a four-sub-score rubric (evidence, logic, relevance, persuasion) plus pairwise comparative output, with quality bands conditioned on format (Policy/LD/PF/Congressional) and circuit (lay/national). Draw structural templates from DebateBench [WEB-28] and CompAQA [WEB-26]. Validate the taxonomy with NSDA coaches and TOC-circuit judges before deployment.

ID	Evidence
WEB-26	CompAQA argues for combined pairwise classification and ranking — the kind of output form the deployment requires and FOLIO lacks (https://www.mdpi.com/2079-9292/13/20/4088)
WEB-28	DebateBench evaluates speech-level scoring and ranking — confirms multi-output debate benchmarks are feasible but FOLIO does not match (https://arxiv.org/pdf/2502.06279)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US High School Competitive Debate — Argument Quality Assessment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO ground-truth labels are mechanically derived from a theorem prover [Q2, Q57] and human annotators are CS undergraduates/graduates with formal FOL training and native or near-native English proficiency [Q27, Q28, Q122, Q123]. The expert/non-expert performance gap (95.98% vs. 61.82%, 34.16 points) confirms that FOL domain knowledge is necessary for the benchmark’s correctness standard [Q107-Q112]. The deployment’s quality-defining community is NSDA/TOC judges, circuit coaches, and the broader competitive-debate community — a categorically different population with different domain expertise and different correctness norms (paradigms, lay vs. tech circuit). Dataset analysis confirms every label is theorem-prover output [DATASET-D9, D12]. The HIGH-priority elicitation flag on OC is fully borne out.

ID	Evidence
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q28	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-” (p.3)
Q57	“Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine.” (p.5)
Q107	“We collected truth value annotations of logical reasoning for FOLIO test set from expert and non-expert annotators.” (p.9)
Q108	“Our expert annotators are computer science college students familiar with FOL.” (p.9)
Q109	“Non-expert annotators are community college or high school students who have not taken the SAT.” (p.9)
Q110	“Both expert and non-expert annotators are native English speakers.” (p.9)
Q111	“Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%.” (p.9)
Q112	“This shows that sufficient domain knowledge of FOL is necessary for good performance on FOLIO.” (p.9)
Q122	“Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.12)
Q123	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing.” (p.12)
D9	$\forall x (\text{InThisTechCompany}(x) \wedge \text{Consistent}(x) \wedge \text{StickTo}(x, \text{theirRegularRoutine}) \rightarrow \neg \text{Like}(x, \text{surprise}))$ — FOL annotation is theorem-prover verified; labels are mechanically derived, not from human judgment of argument quality
D12	$\forall x (\text{Country}(x) \wedge \text{PossibleEffectiveMonetaryPolicy}(x) \rightarrow \text{SuccessfulInflationControl}(x) \wedge \text{StongNationalCurrency}(x))$ — Ground truth derived from FOL inference engine, not from domain expert or debate-community judgment

Strengths

- Within the narrow scope of formal deductive validity, labels are mechanically verified and reproducible [Q2, Q57] — no annotator disagreement or subjective variance
- Annotation process is rigorous: 980 person-hours across six stages [Q32], senior researcher review [Q31], dedicated FOL and NL expert subgroups [Q29, Q124, Q125]

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q31	"All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria." (p.4)
Q32	"We develop our dataset in six stages: WikiLogic collection, HybLogic collection, NL quality control, FOL quality control, NL-FOL alignment and FOL verification, spending 980 man-hours in total." (p.4)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels are theorem-prover outputs [\[Q2, Q57\]](#) derived from FOL annotations by CS-trained annotators [\[Q27, Q28\]](#). They do not reflect debate-community quality judgments — paradigm-based judging, circuit norms, and coach rubrics play no role.

OC-2: The annotator population (CS students familiar with FOL [\[Q108\]](#)) is categorically different from the deployment's quality-defining community (NSDA judges, TOC paradigms, coaches). The non-expert annotation experiment in the paper itself shows that even high school students unfamiliar with FOL achieve only 61.82% accuracy on FOLIO labels [\[Q109, Q111\]](#) — indicating that FOLIO's correctness standard does not converge with the population most similar to the deployment's end users.

OC-3: Annotator demographics are documented: native or near-native English speakers, CS/graduate students with FOL coursework or self-study [\[Q27, Q28, Q122, Q123\]](#). No debate-community annotators are reported.

OC-4: Label re-annotation by debate judges/coaches would not produce comparable labels — the labels capture FOL entailment, not argument quality. Re-annotation would require redefining the task, not refining labels.

OC-5: Labels are deterministic (theorem-prover output), so aggregation does not erase minority perspectives within FOLIO's scope. However, this very determinism is the problem: pluralistic debate-community judgment cannot be approximated by a deterministic prover [\[Q137\]](#).

OC-6: Convergent validity is violated because FOLIO labels do not correlate with the deployment's correctness standard; external validity is violated because performance on FOLIO labels does not generalize to debate-community quality judgments.

ID	Evidence
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Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q27	"Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.3)
Q28	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-" (p.3)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q108	"Our expert annotators are computer science college students familiar with FOL." (p.9)
Q109	"Non-expert annotators are community college or high school students who have not taken the SAT." (p.9)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q112	"This shows that sufficient domain knowledge of FOL is necessary for good performance on FOLIO." (p.9)
Q122	"Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.12)
Q123	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing." (p.12)
Q137	"FOL has no ambiguity while ambiguity can occur at various levels of NLP." (p.13)
WEB-12	NSDA Nationals adjudication and judge paradigms define a community-norm-based quality standard fundamentally different from theorem-prover labels (https://www.aralia.com/helpful-information/guide-to-the-nsda-nationals/)
WEB-26	CompAQA argues for combined pairwise classification and ranking — the kind of output form the deployment requires and FOLIO lacks (https://www.mdpi.com/2079-9292/13/20/4088)

Requires Expert Verification

- Whether any FOLIO labels would be considered 'persuasive' or 'well-warranted' by debate judges if the corresponding NL stories were used as speech material — likely no systematic correlation, but not empirically tested

Remediation

Gap 1: FOLIO labels are theorem-prover outputs from CS/FOL-trained annotators; deployment correctness is defined by NSDA judges, TOC paradigms, and coaching rubrics — a categorically different community.

Recommendation: Recruit a stratified panel of debate-community annotators (NSDA-credentialed judges, active coaches, recent national-circuit competitors) to produce ground-truth scores on the deployment-specific rubric. Stratify by format and by lay vs. tech circuit. Use paradigm metadata from Tabroom to surface circuit-conditioned norms. Measure and report inter-rater agreement; expect lower kappa than FOLIO (consistent with [WEB-26]) and design aggregation to preserve circuit-level signal rather than averaging it away.

ID	Evidence
WEB-26	CompAQA argues for combined pairwise classification and ranking — the kind of output form the deployment requires and FOLIO lacks (https://www.mdpi.com/2079-9292/13/20/4088)

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US High School Competitive Debate — Argument Quality Assessment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's output form is a single categorical label per conclusion [Q80, Q153], with the NL-FOL translation task using Syntactic Validity and Inference Engine Execution Accuracy [Q73-Q76]. The deployment requires multi-dimensional sub-scores, free-text coaching rationales, and pairwise comparative outputs (per elicitation A2). Dataset analysis confirms all 57 examples produce single labels with no sub-scores, no rationales, no comparative structures [DATASET-D13, D14, D19, D21]. The authors themselves acknowledge the translation metric is insufficiently reliable [Q77]. The HIGH-priority elicitation flag on OF is fully borne out.

ID	Evidence
Q73	"Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV)." (p.6)
Q74	"The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2)." (p.6)
Q75	"Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion." (p.6)
Q76	"We define the accuracy of the output labels as the execution accuracy." (p.6)
Q77	"We leave for future work the design of a more reliable metric of NL-FOL translation." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
Q153	"Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises." (p.14)
D13	All professional athletes spend most of their time on sports. All Olympic gold medal winners are professional athletes. ... If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner. — Six-premise chain requiring multi-step deductive inference; output is single True/False/Unknown label
D14	People who like financial risks invest in the public stock market regularly or enjoy gambling regularly. ... Matt does not invest in the public stock market regularly. Matt likes financial risks. — Complex multi-step deduction; conclusion collapses to single categorical label with no sub-scores or rationale
D19	No athletes never exercise. ... Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises. — Negation/exclusive-disjunction complexity; output remains a single binary-equivalent label
D21	Customers in James' family subscribe to AMC A-List or HBO service. ... Lily is in James' family; she watches TV series in cinemas. — Everyday consumer scenario; label is machine-verified True — no qualitative feedback dimension possible

Strengths

- Accuracy over a balanced three-way label space [Q80, Q94] is a clean, reproducible metric within FOLIO's scope
- Multiple random seeds and averaged results [Q93] support statistical reliability of the metric
- Premise-shuffling analysis confirms metric is not gameable via ordering heuristics [Q158-Q160]

ID	Evidence
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)

Q93	"We run experiments on five randomly sampled sets of examples from the training set and report the average accuracy." (p.7)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
Q158	"To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models." (p.15)
Q159	"We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises." (p.15)
Q160	"This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (single categorical label) does not match deployment needs: separate sub-scores per dimension, free-text qualitative feedback, and pairwise rankings are all required [elicitation A2]. None can be derived from FOLIO's output form.

OF-2: Not applicable — deployment is text-based; speech output not required in current scope.

OF-3: Not applicable in the conventional sense — target users are literate US high school students.

OF-4: Form mismatches are fundamental: benchmark performance metrics (accuracy, SynV, ExcAcc) are structurally uninformative for assessing whether a model can produce multi-dimensional argument scores, coaching rationales, or pairwise comparisons. The translation metric is itself flagged as insufficient by the authors [Q77].

ID	Evidence
Q73	"Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV)." (p.6)
Q76	"We define the accuracy of the output labels as the execution accuracy." (p.6)
Q77	"We leave for future work the design of a more reliable metric of NL-FOL translation." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
Q153	"Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises." (p.14)
WEB-26	CompAQA argues for combined pairwise classification and ranking — the kind of output form the deployment requires and FOLIO lacks (https://www.mdpi.com/2079-9292/13/20/4088)
WEB-28	DebateBench evaluates speech-level scoring and ranking — confirms multi-output debate benchmarks are feasible but FOLIO does not match (https://arxiv.org/pdf/2502.06279)

Remediation

Gap 1: FOLIO produces a single categorical label per example; the deployment requires multi-dimensional scores, free-text coaching rationales, and pairwise comparison outputs.

Recommendation: Design the evaluation harness to require structured multi-score output plus a free-text rationale per sub-score, plus a separate pairwise comparison subtask. Evaluate rationale quality with both automated metrics

(faithfulness, citation correctness) and judge-panel review. FOLIO accuracy can be used only as an ancillary diagnostic for deductive-validity behavior, not as a deployment KPI.